

# Relational Schema

|                                                                                           |                                                                                                                                 |                                                                                         |                                                                                                                                                                                    |                                                                                                                                                                                    |
|-------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>DEPARTMENT</b><br><u>dept_id</u> (PK)<br>dept_name                                     | <b>FACULTY</b><br><u>faculty_id</u> (PK)<br>name<br>email<br>is_hod<br>dept_id (FK)                                             | <b>STUDENT</b><br><u>student_id</u> (PK)<br>roll_no<br>name<br>semester<br>dept_id (FK) | <b>COURSE</b><br><u>course_id</u> (PK)<br>course_code<br>credits<br>dept_id (FK)                                                                                                   | <b>ENROLLMENT</b><br><u>student_id</u> (PK, FK)<br><u>course_id</u> (PK, FK)<br>academic_year                                                                                      |
| <b>CLASS_SESSION</b><br><u>session_id</u> (PK)<br>course_id (FK)<br>date<br>time<br>topic | <b>ATTENDANCE</b><br><u>session_id</u> (PK, FK)<br><u>student_id</u> (PK, FK)<br>status<br>(Present / Absent / On Leave / Late) | <b>LEAVE_TYPE</b><br><u>type_id</u> (PK)<br>type_name<br>max_days_per_sem               | <b>LEAVE_REQUEST</b><br><u>request_id</u> (PK)<br>student_id (FK)<br>type_id (FK)<br>from_date<br>to_date<br>reason<br>status<br>(Pending / Mentor Approved / Approved / Rejected) | <b>LEAVE_APPROVAL</b><br><u>approval_id</u> (PK)<br>request_id (FK)<br>level<br>(Mentor / HOD)<br>decision<br>(Approved / Rejected)<br>remarks<br>approval_date<br>faculty_id (FK) |

**Primary Keys (PK)**

- department(dept\_id)
- faculty(faculty\_id)
- student(student\_id)
- course(course\_id)
- class\_session(session\_id)
- leave\_type(type\_id)
- leave\_request(request\_id)
- leave\_approval(approval\_id)
- enrollment(student\_id, course\_id)
- attendance(session\_id, student\_id)

**Foreign Keys (FK)**

- faculty.dept\_id → department.dept\_id
- student.dept\_id → department.dept\_id
- course.dept\_id → department.dept\_id
- enrollment.student\_id → student.student\_id
- enrollment.course\_id → course.course\_id
- class\_session.course\_id → course.course\_id
- attendance.session\_id → class\_session.session\_id
- attendance.student\_id → student.student\_id
- leave\_request.student\_id → student.student\_id
- leave\_request.type\_id → leave\_type.type\_id
- leave\_approval.request\_id → leave\_request.request\_id
- leave\_approval.faculty\_id → faculty.faculty\_id

```
Plain text-
DEPARTMENT(
    dept_id PK,
    dept_name
)

FACULTY(
    faculty_id PK,
    name,
    email,
    is_hod,
    dept_id FK → DEPARTMENT.dept_id
)

STUDENT(
    student_id PK,
    roll_no,
    name,
    semester,
    dept_id FK → DEPARTMENT.dept_id,
    mentor_id FK → FACULTY.faculty_id
)

COURSE(
    course_id PK,
    course_code,
    credits,
    dept_id FK → DEPARTMENT.dept_id
)

ENROLLMENT(
    student_id PK, FK → STUDENT.student_id,
    course_id PK, FK → COURSE.course_id,
    academic_year
)

CLASS_SESSION(
    session_id PK,
    course_id FK → COURSE.course_id,
    date,
    time,
    topic
)
```

```
ATTENDANCE(  
    session_id PK, FK → CLASS_SESSION.session_id,  
    student_id PK, FK → STUDENT.student_id,  
    status  
)
```

```
LEAVE_TYPE(  
    type_id PK,  
    type_name,  
    max_days_per_sem  
)
```

```
LEAVE_REQUEST(  
    request_id PK,  
    student_id FK → STUDENT.student_id,  
    type_id FK → LEAVE_TYPE.type_id,  
    from_date,  
    to_date,  
    reason,  
    status  
)
```

```
LEAVE_APPROVAL(  
    approval_id PK,  
    request_id FK → LEAVE_REQUEST.request_id,  
    faculty_id FK → FACULTY.faculty_id,  
    level,  
    decision,  
    remarks,  
    approval_date  
)
```